

E-Commerce Platform Build Roadmap

A four-phase plan covering structure, backend foundations, core commerce features, and localization & launch — from first skeleton to production handoff.

4

PHASES

100%

CONFIRMATION
TARGET

67–90

ESTIMATED
DAYS

AR/EN

LOCALIZATION

Roadmap Overview

01	Structure & UI Skeleton	7–15 days
02	Backend Foundation & Data Layer	25–30 days
03	Core E-commerce Features	25–30 days
04	Language, SEO & Launch	10–15 days
+	Pro Features — Not Included in Current Budget	Optional add-ons

Each phase below represents roughly one quarter of the total build. Estimated day ranges assume sequential execution; phases 2 and 3 carry the heaviest engineering load, while phase 4 focuses on polish, compliance, and a clean handoff.

Structure & UI Skeleton

Foundational routing, components, and design system

7–15 days

Main Route System

STOREFRONT

- `/` — main store (home)
- `/products` — listing with filter query params
- `/[productId]` — product details
- `/[productId]#[reviewId]` — deep link to a review
- `/cart` , `/checkout` , `/favorites`
- `/search?q=`
- `/login` , `/register`
- `/profile` — order history & saved info
- `/orders/[orderId]` — order tracking
- `/category/[slug]` — category landing
- `/about-us` , `/privacy-policy` , `/terms` , `/contact-us`

ADMIN (PROTECTED)

- `/admin` — dashboard home
- `/admin/login` — separate login screen
- `/admin/products` — list / add / edit
- `/admin/products/[productId]/edit`
- `/admin/orders` , `/admin/orders/[orderId]`
- `/admin/customers`
- `/admin/coupons`
- `/admin/reports` — sales & performance
- `/admin/settings` — SEO fields, store settings

Main Components

- **User NavBar** — logo, search, cart, favorites, login/profile, language switcher
- **User Footer** — page links, social, contact, copyright
- **Admin NavBar** — store name, employee name, notifications, logout
- **Admin Sidebar** — Products / Orders / Customers / Coupons / Reports / Settings
- **Admin Footer**
- **Filter** — category / price / color / size
- **Search Bar** — with live suggestions
- **Product Card** — image, name, price, stock badge, quick favorite
- **Product Grid / List** — pagination or infinite scroll
- **Breadcrumbs & Pagination**
- **Language Switcher** (AR / EN)
- **Toast, Loading Skeletons, Admin Data Table**

Design System Setup

- Import brand identity from company profile: logo, primary blue palette, typography
- Tailwind config: custom colors, RTL-aware spacing/utility classes
- Base UI kit — buttons, inputs, badges, modals, cards — built once, reused everywhere

Backend Foundation & Data Layer

Database schema, API structure, and authentication

25–30 days

Database Schema (PostgreSQL)

Table	Key Fields
users	id, name, email, hashed_password, oauth_provider, oauth_id, role
categories	id, name_ar, name_en, slug, parent_id
products	id, name_ar/en, description_ar/en, price, sku, category_id, is_available, meta_title, meta_description, slug
product_variants	id, product_id, size, color, stock_quantity
product_images	id, product_id, image_url, sort_order
carts / cart_items	supports guest session-based and logged-in carts
favorites	user_id, product_id
orders	id, user_id/guest_info, status, payment_method/status, total, shipping_fee, coupon_id
order_items	order_id, product_variant_id, quantity, price_at_purchase
coupons	code, discount_type, discount_value, expiry_date, usage_limit
reviews	id, product_id, user_id, rating, comment, created_at
shipments	order_id, courier_name, tracking_number, status

FastAPI Project Structure

- `/app/routers` — auth, products, categories, cart, orders, coupons, reviews, admin
- `/app/models` — SQLAlchemy models
- `/app/schemas` — Pydantic request/response models
- `/app/services` — pricing, stock checks, coupon validation
- `/app/core` — config, security/hashing, JWT
- Alembic migrations initialized & version-controlled

Auth System

- Email/password registration & login (bcrypt/argon2 hashing)
- Google OAuth login
- Facebook OAuth login
- JWT-based session for logged-in users
- Guest checkout — Redis-backed session/cookie cart
- `role` field reserved for future permission splits

Core API Endpoints (First Pass)

Auth: register / login / oauth / me

Products: list / filter / detail / search

Categories: list / detail

Cart: get / add / update / remove

Redis: guest sessions + caching

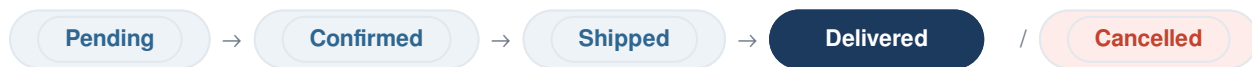
Core E-commerce Features

Checkout, discovery, admin tools, and shipping

25–30 days

Checkout & Payments

- Kashier integration for online payment (cards, wallets) — EGP only
- Cash on Delivery (COD) as a manual method with its own status flow
- Shipping fee calculation — no tax logic, per requirements
- Order confirmation triggers: in-site notification + transactional email (Resend)



Shopping & Discovery

- Coupon system — apply/validate at checkout, admin CRUD
- Search & filters wired end-to-end (price / category / color / size)
- Favorites / wishlist — persisted per logged-in user
- Out-of-stock products shown, clearly badged
- Product reviews — rating + comment, linked via `#[reviewId]`

Admin Dashboard Features

- **Product CRUD** — name, description, price, images, category, variants
- **Order management** — view, update status, view customer info
- **Customer management** — registered customers & order history
- **Coupon management** — create / edit / deactivate
- **Reports** — best sellers, order count, total sales, monthly reports (charts + tables)
- **Notification center** — order & low-stock alerts, dashboard-internal only

Shipping integration — confirm courier with client (e.g. Bosta, Mylerz) and integrate their tracking API. Status reflected on both `/orders/[orderId]` and `/admin/orders/[orderId]`.

Language, SEO & Launch

Localization, performance, testing, and deployment

10–15 days

Language

- Localization system using **next-intl**
- Arabic (default) and English, full RTL/LTR switching
- All static copy, product fields, and admin content localized

SEO

- Editable meta title & description per page/product
- Clean, human-readable URL slugs
- Auto-generated **sitemap.xml** & **robots.txt**
- Structured data (Product schema) for product pages

Static Content Pages

- About Us, Privacy Policy, Terms, Contact Us — real company content

Performance & Hardening

- Image optimization via **next/image**, served from Cloudflare R2
- API response caching (Redis) for listings/categories
- Rate limiting on auth & checkout endpoints

Testing

- Backend unit tests (pytest) — pricing, coupon, stock logic
- End-to-end tests — guest checkout, login checkout, admin flows

Deployment

- Docker Compose — Next.js, FastAPI, PostgreSQL, Redis, Nginx on Hostinger VPS
- Nginx reverse proxy — **/** → Next.js, **/api** → FastAPI
- SSL via Hostinger's built-in SSL
- Domain connected once client confirms ownership / registration

Monitoring, backups & handoff — automated daily PostgreSQL backups, basic uptime monitoring, final QA pass against every requirement in the clarification document, plus a short training session and handoff doc for the admin employee.



OPTIONAL ADD-ONS

Main Pro Features

The following are not included in the current budget. They can be scoped and added as extensions once the core platform above is delivered.

Database Optimization Partially included in current scope	~3,000–5,000 EGP
Performance Live Tracker Real-time performance monitoring dashboard	~2,000–3,000 EGP
Authentication Live Tracker Optimizing security monitoring for auth flows	~2,000–3,000 EGP
More Themes Current budget includes one light theme only	~1,000 EGP

Ready to build.

Each phase closes with a QA checkpoint against the client's confirmed requirements before moving forward.